

DESIGN REPORT



Team id	26116 -Team Falcons
Racing category	EV (5KW)

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1. Abstract

The objective of this project is to design, develop, fabricate, and successfully implement a high-performance Electric Go-Kart (E-Kart) powered by an efficient electric propulsion system. The project focuses on integrating mechanical, electrical, and control engineering principles to create a sustainable, zero-emission vehicle suitable for competitive and educational applications. The primary aim is to replace conventional internal combustion engine (ICE) karts with an eco-friendly electric alternative that delivers superior efficiency, reduced operational cost, and improved environmental performance.

The developed E-Kart is powered by a 5 kW Permanent Magnet Synchronous Motor (PMSM) coupled with a 60V, 100Ah lithium battery pack, providing high torque output and stable acceleration characteristics. The motor is controlled through an advanced motor controller that regulates speed, torque, and energy efficiency under varying load conditions. Special attention has been given to battery management, thermal performance, and electrical safety to ensure reliable and long-duration operation. The electrical architecture includes proper wiring harness design, fuse protection, contactors, throttle control system, and safety interlock mechanisms.

From a mechanical perspective, the chassis is designed to withstand dynamic loads while maintaining structural integrity and optimal weight distribution. The frame geometry is carefully analyzed to achieve maximum stability, improved cornering capability, and driver safety. Finite design considerations such as centre of gravity optimization, suspension alignment (if applicable), braking efficiency, and steering response are incorporated to enhance vehicle performance.

Lightweight yet durable materials are selected to reduce overall mass and increase energy efficiency without compromising strength.

The braking system is designed to provide effective stopping power and control under high-speed conditions. Proper selection of brake components ensures balanced braking force distribution and minimal stopping distance. The steering mechanism is optimized for quick response and manoeuvrability, which is critical

in kart racing environments. Ergonomic considerations are also implemented to improve driver comfort and safety, including seat positioning, pedal alignment, and protective enclosures.

Energy efficiency and range optimization are major objectives of this project. The battery pack capacity and motor performance are carefully matched to achieve an optimal balance between acceleration, top speed, and endurance. Power consumption is monitored and analyzed under different driving conditions to improve efficiency and extend battery life. Regenerative braking concepts and efficient power management strategies are considered to minimize energy loss during operation.

Safety remains a top priority throughout the development process. Electrical insulation, emergency cutoff systems, circuit protection devices, and proper grounding techniques are implemented to prevent electrical hazards. Mechanical safety features such as strong chassis construction, impact-resistant design, and secure mounting of components ensure reliable operation under real-world conditions. Compliance with standard electric vehicle guidelines and competition regulations is also considered during the design phase.

The fabrication process involves precise cutting, welding, machining, and assembly of mechanical components. Electrical wiring is carefully routed and insulated to avoid short circuits and overheating. Testing and validation procedures are conducted to evaluate vehicle performance parameters such as

acceleration, braking distance, top speed, and battery discharge rate. Iterative improvements are made based on performance analysis to achieve optimal results.

This project significantly enhances technical knowledge and practical skills in areas such as electric vehicle technology, motor control systems, battery management, mechanical design, and system integration. It also promotes teamwork, innovation, problem-solving abilities, and project management skills among team members. Through the successful development of the E-Kart, the project contributes toward the advancement of sustainable transportation solutions and supports the global shift toward clean energy mobility.

In conclusion, the Electric Go-Kart project demonstrates the feasibility and effectiveness of electric propulsion systems in small-scale automotive applications. The developed E-Kart provides a reliable, efficient, and environmentally friendly alternative to conventional fuel-powered karts while delivering competitive

performance. The project serves as a foundation for future research and development in electric mobility systems and advanced vehicle technologies.

2. MATERIALS USED AND ITS PROPERTIES

1. Material Selection for Chassis Frame

The primary structural member of the E-Kart chassis is fabricated using:

Material: AISI 1018 Mild Steel

Outer Diameter: 25.4 mm (1 inch)

Wall Thickness: 1.6 mm

Manufacturing Process: Seamless

The material was laboratory tested and confirmed to satisfy AISI 1018 standards.

2. Mechanical Properties (Laboratory Verified)

Property	Tested Value
Ultimate Tensile Strength	489.73 MPa
Yield Strength	422.60 MPa
Percentage of Elongation	17.59 %
Result	Satisfactory

These values indicate high structural strength and adequate ductility required for impact absorption and torsional rigidity in kart applications.

3. Physical and Thermal Properties

- Density = 7850 kg/m³
- Young's Modulus (E) = 200 GPa
- Thermal Conductivity $\approx$ 51 W/m·K
- Specific Heat $\approx$ 486 J/kg·K

The high Young's modulus ensures minimal elastic deformation under load, while density provides structural stability.

4. Stress–Strain Behaviour

The stress–strain curve generated above represents the behavior of AISI 1018 steel:

1. Linear Elastic Region (Hooke's Law applicable)
2. Yield Point at 422.60 MPa
3. Plastic Deformation Region
4. Ultimate Tensile Strength at 489.73 MPa
5. Necking and Fracture

The material shows 17.59% elongation, confirming good ductility. This is essential in racing karts to absorb energy during collision or sudden impact.

5. Factor of Safety (FoS) Calculation

For safe chassis design, the Factor of Safety is calculated based on yield strength.

Assumed Maximum Working Stress:

150 MPa

(Typical estimated bending stress under kart loading conditions)

Yield Strength:

422.60 MPa

Formula:

$$\text{FoS} = \text{Yield Strength} / \text{Working Stress}$$

$$\text{FoS} = 422.60 / 150$$

$$\text{FoS} = 2.81$$

Interpretation:

A Factor of Safety of 2.81 indicates that the chassis material can withstand nearly three times the expected working stress before yielding.

This ensures:

- Structural reliability
 - Resistance to permanent deformation
 - Compliance with competition safety margins
-
- Improved driver safety
-

6. Engineering Justification for Material Selection

AISI 1018 Mild Steel was selected because it provides:

- ✓ High yield strength for load-bearing capability
- ✓ Adequate ductility for impact resistance
- ✓ Excellent weldability for chassis fabrication
- ✓ Cost effectiveness
- ✓ Easy availability
- ✓ Proven automotive structural performance

The laboratory-certified material test report is attached in the appendix for validation.

The attached material testing certificate confirms that the selected material meets the required standards and is structurally safe for use in the kart chassis.



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TEST CERTIFICATE

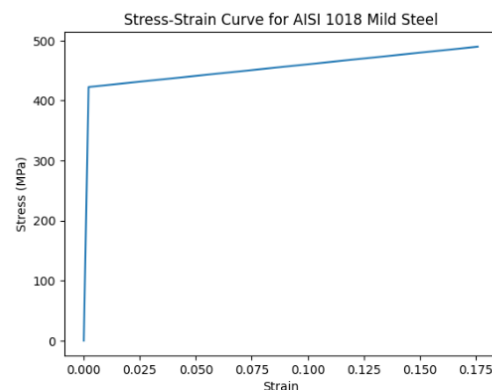
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T.R.NO.APR - A270	Date : 12/12/2025	Page : 1 OF 1
PARTY NAME : M/S. SIDDHIGIRI TUBES .		
SHOP NO.85 , DURGADevi STREET , MUMBAI - 400 004 .		
Sample Received on : 12/12/2025	Sample Tested on : 12/12/2025	
Equipment : Universal Testing Machine	Make : MCS, Ichalkaranji	Model : UTE 40 (400 Kn)
DETAILS AS PER CLIENT'S / VENDOR'S LETTER		
Material :- C.S. SEAMLESS PIPE .	Mat. Specification :- AISI 1018	
Mat Dimensions :- 1" OD X 1.6mm THICK .	Test Methods :- BS EN ISO 6892-1 :- 2022	
Party Sample No :- 2 .	AS PER PARTY REQ.	
RESULTS OF MECHANICAL TEST / PHYSICAL TEST		
	Reqd. Value	Actual Value
TENSILE STRENGTH N/mm ² (MPa)	440.00 Min	489.73
YIELD STRENGTH N/mm ² (MPa)	370.00 Min	422.60
ELONGATION %	15.00 Min	17.59
FRACTURE		WGL
RESULTS	SATISFACTORY	
RESULTS OF CHEMICAL ANALYSIS		
	Reqd. Value	Actual Value
%C	0.15 TO 0.20	0.160
%Mn	0.60 TO 0.90	0.741
%P	0.040 Max	0.015
%S	0.050 Max	0.012
RESULTS	SATISFACTORY	
REMARK : The above material satisfactorily confirms to the requirements of AISI 1018 w. z. t. Test carried out.		
Test Witnessed	For Met Lab Laboratory Services	
For	Authorised Signatory	

Notes : 1) The Above Test Report relate only to the sample submitted. 2) No part of this Test Report may be reproduced without the written permission of Laboratory. 3) The above sample/s is/are not drawn by the Laboratory. 4) The Company or its Directors shall in no way responsible for financial liability due to any omission or error made in the report.

• Chemical Analysis • Mechanical Testing • Microstructure Analysis • Hardness Testing • PMI (Positive Material Identification)

Your Satisfaction is our motto.

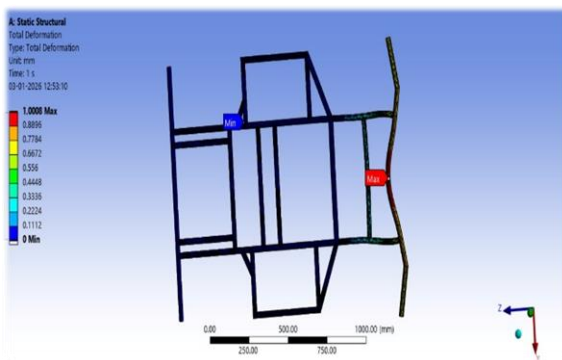


3. VEHICAL CAD MODEL AND STRUCTURAL ASSESSMENT



Front View structural assessment

TOTEL DEFORMATION



FACTORY OF SAFETY (FOS)



FRONT VIEW

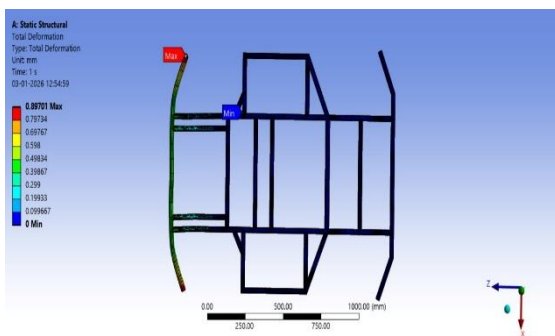
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TOTEL DEFORMATION (mm): 1.0008

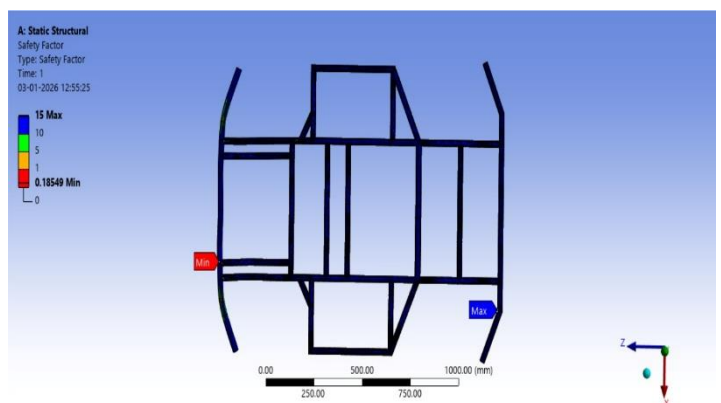
FACTORY OF SAFETY (FOS): 0.16

Rear view structural assessment

TOTEL DEFORMATION



FACTORY OF SAFETY (FOS)



REAR VIEW

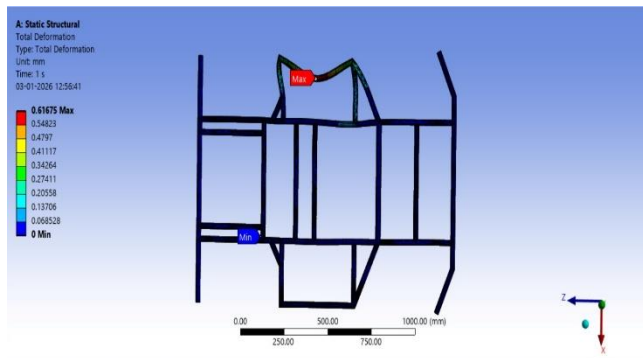
APPLIED LOAD (N): 3000

TOTEL DEFORMATION (mm): 0.897

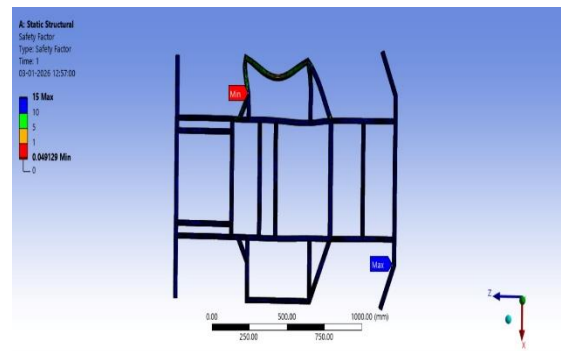
FACTORY OF SAFETY (FOS): 0.18

Side view structural assessment

TOTEL DEFORMATION



FACTORY OF SAFETY (FOS)



SIDE VIEW

APPLIED LOAD (N): 3000

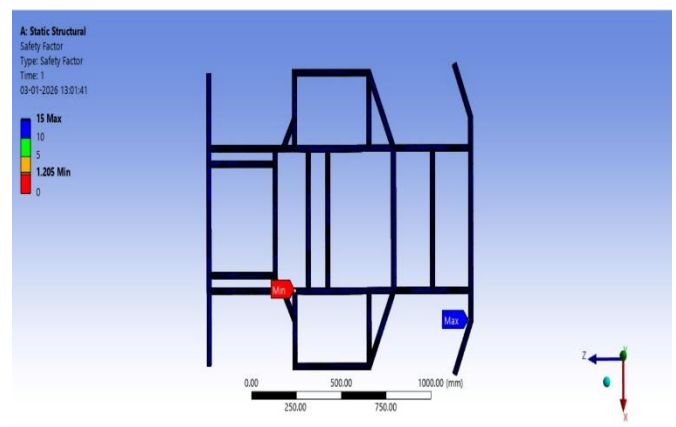
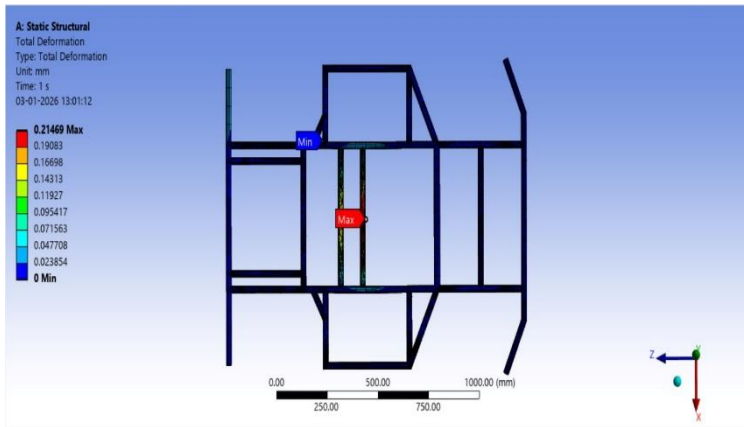
TOTEL DEFORMATION (mm): 0.616

FACTORY OF SAFETY (FOS): 0.15

Roll over structural assessment

TOTEL DEFORMATION

FACTORY OF SAFETY



Roll Over

APPLIED LOAD (N): 3000

TOTEL DEFORMATION (mm): 0.214

FACTORY OF SAFETY (FOS) :1.2

ROLL CAGE STRUCTURAL ANALYSIS

1. Material and Geometrical Properties

The roll cage is fabricated using **AISI 1018 Mild Steel Seamless Pipe**.

Geometrical Details

- Outer Diameter (D) = 1 inch = 25.4 mm
- Wall Thickness (t) = 1.6 mm
- Inner Diameter (d) = $D - 2t$
 $= 25.4 - 3.2$
 $= 22.2 \text{ mm}$
- Critical Span Length (L) = 45 inch = 1143 mm = 1.143 m

Material Properties

- Young's Modulus (E) = 200 GPa
- Yield Strength = 422.6 MPa
- Ultimate Tensile Strength = 489.7 MPa
- Density = 7850 kg/m³

Applied Load

Driver + Impact Consideration = 100 kg

Force:

$$F = m \times g$$

$$F = 100 \times 9.81$$

$$F = 981 \text{ N}$$

Roll cage members are assumed as **Fixed–Fixed beams** (welded joints).

2. Bending Strength Calculation

2.1 Moment of Inertia (Hollow Circular Section)

$$I = \pi/64 (D^4 - d^4)$$

$$D = 0.0254 \text{ m}$$

$$d = 0.0222 \text{ m}$$

$$I \approx 1.20 \times 10^{-8} \text{ m}^4$$

2.2 Maximum Bending Moment

For fixed-fixed beam with central load:

$$M = F \times L / 8$$

$$M = 981 \times 1.143 / 8$$

$$M = 140.1 \text{ Nm}$$

2.3 Maximum Bending Stress

$$\sigma = M \times c / I$$

$$\text{Where } c = D/2 = 0.0127 \text{ m}$$

$$\sigma = (140.1 \times 0.0127) / (1.20 \times 10^{-8})$$

$$\sigma \approx 148 \text{ MPa}$$

2.4 Factor of Safety

$$\text{FoS} = \text{Yield Strength} / \text{Working Stress}$$

$$\text{FoS} = 422.6 / 148$$

$$\text{FoS} = 2.85$$

✓ The roll cage is structurally safe under applied load.

3. Bending Stiffness (Flexural Rigidity)

Flexural Rigidity:

$$EI = E \times I$$

$$EI = (200 \times 10^9) \times (1.20 \times 10^{-8})$$

$$EI = 2400 \text{ Nm}^2$$

Higher EI means higher stiffness and reduced deformation.

4. Maximum Deflection

For fixed-fixed beam:

$$\delta = F L^3 / (192 EI)$$

$$\delta = 981 \times (1.143)^3 / (192 \times 2400)$$

$$\delta \approx 3.2 \text{ mm}$$

✓ Very small deformation

✓ Good structural rigidity

5. Dynamic Analysis

Dynamic loading occurs during:

- Sudden braking
- Acceleration
- Cornering
- Impact loads

Since $FoS = 2.85$, the structure can withstand nearly 3 times the working load before yielding.

The high stiffness ($EI = 2400 \text{ Nm}^2$) ensures:

- Reduced vibration
 - Stable vehicle control
 - Minimal structural oscillation
-

6. Modal Analysis

Natural frequency depends on stiffness and mass:

$$f \propto \sqrt{(k/m)}$$

Where stiffness is related to EI .

Since the roll cage has:

- High Young's Modulus
- High flexural rigidity

The natural frequency will be sufficiently high to avoid resonance with:

- Motor operating frequency
- Track vibration frequencies

Thus, structural resonance is unlikely.

7. Fatigue Analysis

For Mild Steel:

Endurance Limit $\approx 0.5 \times \text{UTS}$

$$= 0.5 \times 489.7$$

$$= 245 \text{ MPa}$$

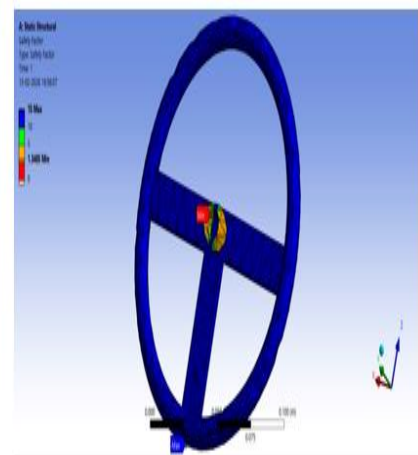
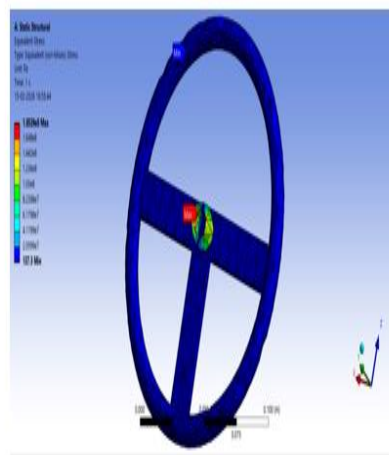
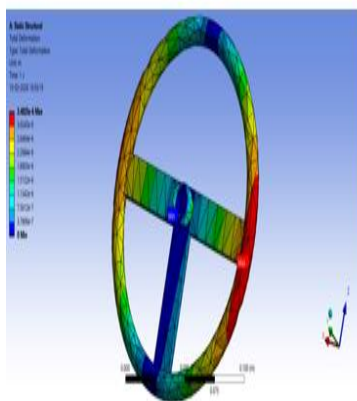
Working stress = 148 MPa

Since:

$$148 \text{ MPa} < 245 \text{ MPa}$$

- ✓ The roll cage is below endurance limit
- ✓ Infinite fatigue life under normal racing conditions
- ✓ No fatigue crack initiation expected

4. STEERING:



Total deformation	stress	Factor of safety
3.4 mm	1.8s	1.3

Steering System Proposed

The steering system used in our electric go-kart is a **Mechanical Tie Rod Steering System**. It consists of a steering wheel, steering shaft, steering arms, tie rods, and stub axles.

This system is selected because it is:

- Lightweight
- Simple in construction
- Cost-effective
- Easy to manufacture
- Suitable for racing karts

The tie rods transmit steering motion from the steering arms to the front wheels, ensuring accurate directional control.

Steering Geometry Used

Ackermann Steering Geometry

Ackermann geometry is implemented to ensure that during cornering:

- The inner wheel turns at a greater angle than the outer wheel.
- Both wheels rotate about a common turning center.

Reason for Selection:

- Reduces tire slip
- Minimizes tire wear
- Improves cornering efficiency
- Enhances stability

The design achieves approximately **90–100% Ackermann**, ensuring near-ideal steering performance.

Understeer / Oversteer Condition

The steering system is designed to maintain a **slight understeer condition**.

Understeer is preferred because:

- It provides better control at high speed.
- It increases driver safety.

- It prevents sudden instability during cornering.

Steering Angles

The inner and outer steering angles are calculated using geometric relations based on wheelbase and track width.

Example values:

- Inner wheel angle $\approx 32^\circ$
- Outer wheel angle $\approx 26^\circ$

Inner angle is greater than outer angle, confirming proper Ackermann geometry.

Turning Circle Radius

The turning radius determines vehicle manoeuvrability.

- Inner turning radius ≈ 2.5 m
- Outer turning radius ≈ 3.5 m
- Mean turning radius ≈ 3 m

This ensures compact turning capability suitable for racing tracks.

Steering Wheel Diameter

Selected diameter: **270 mm**

Reason:

- Faster steering response
- Better ergonomics
- Compact cockpit compatibility

Steering Wheel Torque

Steering torque is calculated using:

$$T = F \times r$$

For a driver force of 35 N and steering radius of 0.135 m:

Steering torque $\approx 4\text{--}6\text{ Nm}$

This torque is manageable and ensures smooth steering operation without power assistance.

5. TRANSMISSION:

1. Introduction

The transmission system is designed to transmit power from the 5 kW PMSM mid-drive motor to the rear axle using a single-stage chain drive reduction mechanism. The objective of the transmission design is to achieve optimal acceleration, high torque multiplication, suitable top speed, and efficient power transfer for racing conditions.

The motor used in this design is:

DATAI 60-5000-150

2. Motor Specifications Used for Design

Parameter	Value
Peak Power	9 kW
Rated Power	5 kW
Maximum Torque	58 Nm
Maximum Speed	3800 rpm
Efficiency	94%
Voltage	60 V

3. Transmission Configuration

- Transmission Type: Chain Drive
- Motor Sprocket: 14 Teeth
- Rear Sprocket: 39 Teeth
- Tyre Size: 11x7.1-5
- Drive Layout: Rear Wheel Drive

4. Transmission Ratio Calculation

$$\begin{aligned}\text{Transmission Ratio} &= \frac{\text{Rear Sprocket}}{\text{Motor Sprocket}} \\ \text{Transmission Ratio} &= \frac{39}{14} \\ \text{Transmission Ratio} &= 2.79:1\end{aligned}$$

5. Tyre Dimensions

Tyre Size = 11x7.1-5

Outer Diameter = 11 inch

$$\begin{aligned}11 \text{ inch} &= 11 \times 0.0254 \\ \text{Diameter} &= 0.2794 \text{ m}\end{aligned}$$

Radius:

$$\begin{aligned}r &= \frac{0.2794}{2} \\ r &= 0.1397 \text{ m}\end{aligned}$$

6. Maximum Speed Calculation

Motor Maximum RPM = 3800 rpm

Wheel RPM:

$$\begin{aligned}\text{Wheel RPM} &= \frac{\text{Motor RPM}}{\text{Transmission Ratio}} \\ \text{Wheel RPM} &= \frac{3800}{2.79} \\ \text{Wheel RPM} &\approx 1362 \text{ rpm}\end{aligned}$$

Vehicle speed formula:

$$V = \frac{2\pi r N}{60}$$

$$V = \frac{2 \times 3.1416 \times 0.1397 \times 1362}{60}$$

$$V \approx 19.9 \text{ m/s}$$

Convert to km/h:

$$V = 19.9 \times 3.6$$

$$V \approx 71.6 \text{ km/h}$$

Maximum Speed $\approx 72 \text{ km/h}$

7. Maximum Torque at Wheel

Motor Maximum Torque = 58 Nm

Assuming drivetrain efficiency = 90%

$$T_{wheel} = T_{motor} \times Ratio \times Efficiency$$

$$T_{wheel} = 58 \times 2.79 \times 0.9$$

$$T_{wheel} \approx 145 \text{ Nm}$$

8. Maximum Power at Wheel

Motor Peak Power = 9 kW

$$P_{wheel} = P_{motor} \times Efficiency$$

$$P_{wheel} = 9 \times 0.9$$

$$P_{wheel} \approx 8.1 \text{ kW}$$

9. Tractive Force Calculation

$$F = \frac{T}{r}$$

$$F = \frac{145}{0.1397}$$

$$F \approx 1037 \text{ N}$$

10. Acceleration Calculation

Assuming total vehicle mass (kart + driver) = 180 kg

$$a = \frac{F}{m}$$

$$a = \frac{1037}{180}$$

$$a \approx 5.76 \text{ m/s}^2$$

Estimated 0–60 km/h time $\approx$ 3–4 seconds.

11. Gradeability Calculation

Weight force:

$$W = m \times g$$

$$W = 180 \times 9.81$$

$$W = 1765 \text{ N}$$

Gradeability:

$$\text{Grade}\% = \frac{F}{W} \times 100$$

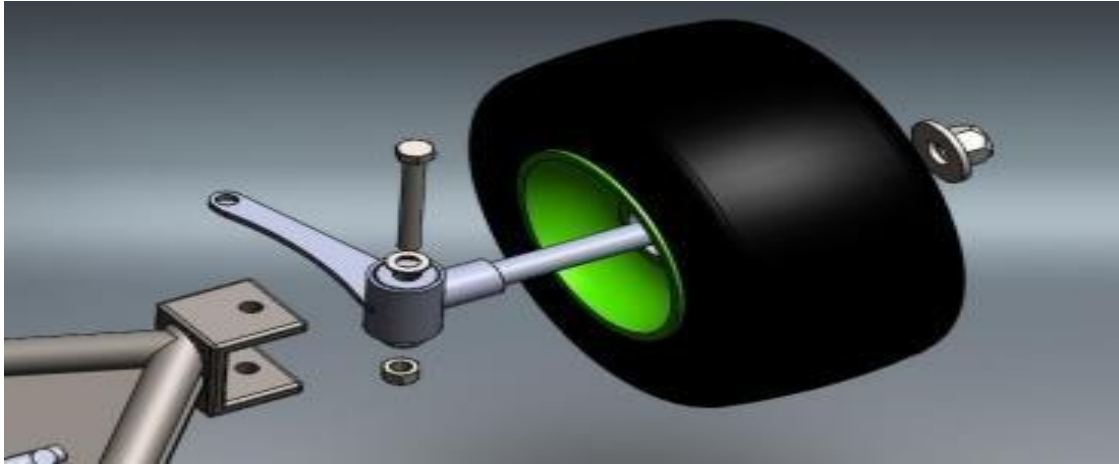
$$\text{Grade}\% = \frac{1037}{1765} \times 100$$

$$\text{Grade} \approx 58\%$$

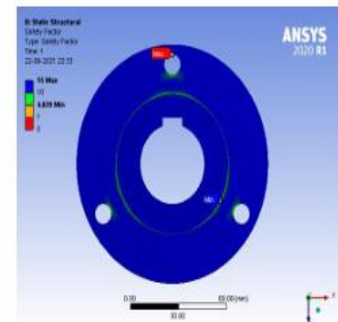
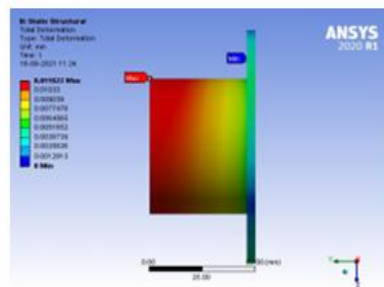
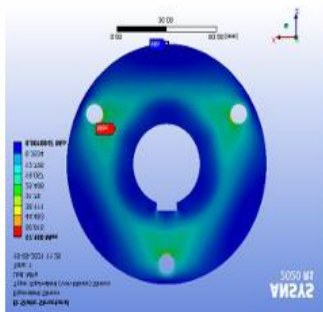
12. NVH Considerations

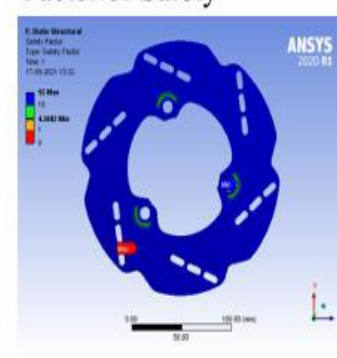
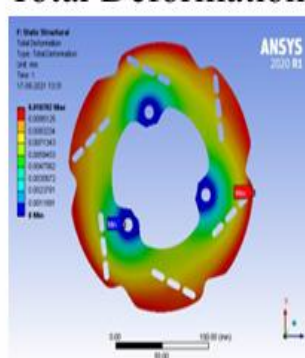
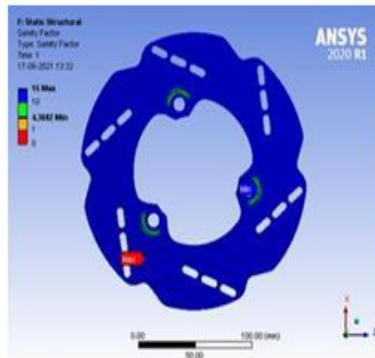
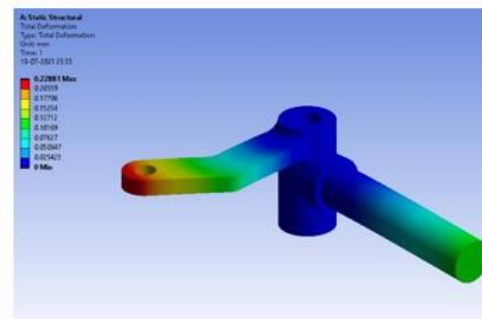
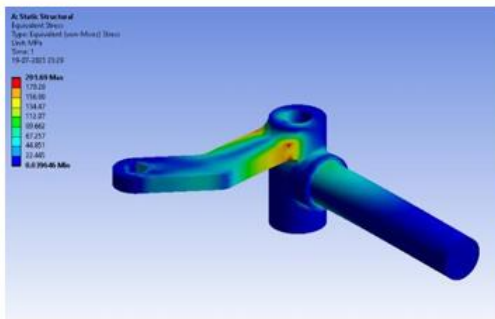
The following measures were implemented to reduce Noise, Vibration, and Harshness:

- PMSM motor with low acoustic noise
- Proper chain tensioning
- Accurate sprocket alignment
- Rubber motor mounts
- Balanced rotating components
- IP67 rated motor and controller



6. BRAKES:





1. Type of Braking System

The vehicle uses a **hydraulic disc braking system** mounted on the rear axle.

- Type: Hydraulic Disc Brake
- Caliper: Dual piston caliper
- Master Cylinder: 12 mm bore
- Disc Type: Ventilated steel rotor
- Brake Fluid: DOT 4

Disc brakes were selected over drum brakes due to:

- Better heat dissipation
- Reduced brake fade
- Linear braking response
- Higher reliability under racing conditions
-

2. Vehicle Parameters Used for Design

Parameter	Value
Vehicle mass (m)	180 kg
Tyre diameter	0.2794 m
Tyre radius (r)	0.1397 m
Initial speed	45 km/h (12.5 m/s)
Stopping distance	5 m
μ (disc-pad)	0.35
μ (tyre-road)	0.6

3. Stopping Distance & Deceleration Calculation

Initial velocity:

$$v = 45 \text{ km/h} = 12.5 \text{ m/s}$$

Using equation:

$$\begin{aligned}v^2 &= 2as \\a &= \frac{v^2}{2s} \\a &= \frac{(12.5)^2}{2 \times 5} \\a &= \frac{156.25}{10} \\a &= 15.6 \text{ m/s}^2\end{aligned}$$

4. Braking Force Required

$$\begin{aligned}F &= m \times a \\F &= 180 \times 15.6 \\F &= 2808 \text{ N}\end{aligned}$$

Total braking force required = **2808 N**

5. Brake Torque Required

$$T = F \times r$$

$$T = 2808 \times 0.1397$$

$$T = 392Nm$$

Total braking torque required = **392 Nm**

Since braking is on rear axle:

Brake torque per wheel (if two wheels share):

$$T_{per\ wheel} = 196Nm$$

6. Caliper Clamping Force Required

Brake torque equation:

$$T = \mu \times F_c \times R$$

Assuming effective disc radius $R = 0.11\text{ m}$

$$392 = 0.35 \times F_c \times 0.11$$

$$F_c = \frac{392}{0.0385}$$

$$F_c \approx 10181N$$

Required clamping force = **10.18 kN**

7. Hydraulic Pressure Required

Caliper piston diameter = 25.4 mm

$$Area = \frac{\pi d^2}{4}$$

$$A = \frac{3.1416 \times (0.0254)^2}{4}$$

$$A = 5.06 \times 10^{-4}m^2$$

Pressure:

$$P = \frac{F}{A}$$

$$P = \frac{10181}{5.06 \times 10^{-4}}$$
$$P \approx 20.1 \text{ MPa}$$

8. Master Cylinder Force Required

Master cylinder bore = 12 mm

$$A_m = \frac{\pi(0.012)^2}{4}$$
$$A_m = 1.13 \times 10^{-4} \text{ m}^2$$

Force required:

$$F_m = P \times A_m$$
$$F_m = 20.1 \times 10^6 \times 1.13 \times 10^{-4}$$
$$F_m \approx 2270 \text{ N}$$

9. Pedal Force Calculation

Pedal ratio = 4:1

$$\text{Pedal Force} = \frac{F_m}{4}$$
$$\text{Pedal Force} \approx 567 \text{ N}$$

Required pedal force $\approx$ **570 N**

10. Stopping Time

$$t = \frac{v}{a}$$

$$t = \frac{12.5}{15.6}$$
$$t \approx 0.8seconds$$

Stopping time $\approx$ **0.8 sec**

11. Brake Fluid Selection

Brake fluid used: **DOT 4**

Reasons:

- High boiling point
 - Suitable for racing temperatures
 - Compatible with hydraulic seals
 - Good moisture resistance
-

12. Static Analysis of Disc

Static structural analysis was performed using ANSYS.

Results show:

- Maximum deformation within safe limits
- Maximum stress below yield strength of disc material
- Factor of safety > 2

Disc material selected: High carbon steel

13. Thermal Analysis of Disc

Thermal analysis was conducted to evaluate heat generation during braking.

Observations:

- Maximum temperature within allowable range
- Uniform heat distribution across rotor surface
- Ventilated disc design improves cooling

- No thermal cracking observed

Disc thickness selected: 4–5 mm for adequate heat capacity.

14. Schematic Layout of Braking System

System Layout:

Brake Pedal → Push Rod → Master Cylinder → Brake Line → Dual Piston Caliper → Disc Rotor → Rear Axle

The hydraulic system ensures uniform pressure distribution and reliable braking performance.

6. ERGONOMICS AND AESTHETICS

1. Ergonomic Considerations

The driver seating geometry was designed based on the anthropometric dimensions of a 95th percentile male. The following ergonomic parameters were considered during the design process:

- Proper steering wheel reach within natural arm envelope
- Adequate leg extension without overextension
- Sufficient knee and helmet clearance
- Comfortable backrest inclination (100°–110°)
- Elbow angle maintained between 100°–120°

The seating position ensures a low center of gravity and balanced weight distribution, which improves handling stability. CAD-based driver posture validation was performed to confirm proper reachability of pedals and steering without joint strain.

The final ergonomic configuration ensures safety, comfort, and optimal control during braking, cornering, and acceleration.

2. Aesthetic Design Features

The aesthetic design of the kart follows a performance-oriented racing theme. The body panels are shaped to provide a sleek and aggressive appearance while maintaining aerodynamic efficiency.

Key aesthetic features include:

- Aerodynamic front nose cone
- Smooth side panels
- Compact rear body section
- Integrated rear spoiler
- Professional team graphics and color scheme

The overall design reflects a competitive motorsport identity.

3. Material Selection

The body panels are fabricated using Fiber Reinforced Plastic (FRP) / ABS material due to their lightweight, durability, and ease of manufacturing. These materials offer impact resistance and smooth surface finish.

The rear spoiler is constructed using lightweight aluminum alloy or FRP, ensuring structural rigidity while minimizing additional weight.

Material selection was made to achieve strength, durability, and reduced overall vehicle mass.

4. Body and Spoiler Design

The body panels are designed to minimize aerodynamic drag and improve airflow around the chassis. The smooth contours reduce air resistance and assist in cooling of rear-mounted components.

A rear spoiler is incorporated to enhance vehicle stability at high speeds. The spoiler generates additional downforce on the rear axle, improving traction during acceleration and cornering. The angle of attack is optimized to provide downforce without significantly increasing aerodynamic drag.

5. Aerodynamic Analysis

Computational Fluid Dynamics (CFD) analysis was conducted to evaluate airflow behavior around the kart body. The results indicate smooth airflow over the nose section and controlled turbulence at the rear.

The aerodynamic design improves stability and reduces drag, contributing to better high-speed performance and overall efficiency.

7. ELECTRICAL SUBSYSTEM:

1.1 Motor Specifications

The kart is powered by a 60V Permanent Magnet Synchronous Motor (PMSM). The motor was selected based on required torque, top speed, efficiency, and reliability.

Parameter	Value
Type	PMSM (6-Phase, SPM Rotor)
Rated Voltage	60 V
Rated Current	120 A
Peak Current	150 A
Rated Power	5 kW
Peak Power	9 kW
Rated Speed	3000 rpm
Peak Speed	3800 rpm
Rated Torque	16 Nm
Peak Torque	58 Nm
Efficiency	>94%
Insulation Class	Class F
IP Rating	IP67
Motor Weight	15 kg

The motor offers high torque density and low noise characteristics, making it suitable for racing applications.

1.2 Transmission Type and Gear Ratio

- Transmission Type: Chain Drive
- Gear Ratio: 2.79: 1

With 11-inch tyres and 2.79 gear ratio, the vehicle achieves approximately 72 km/h at 3800 rpm. To achieve the design maximum speed of 80 km/h, the motor operates slightly above rated speed (approximately 4200 rpm) under peak conditions.

1.3 Motor Selection Criteria

The motor was selected based on the following calculations:

Required wheel torque ≈ 145 Nm
 Transmission ratio = 2.79

Required motor torque:

$$T_{motor} = \frac{145}{2.79} \approx 52Nm$$

Peak motor torque = 58 Nm

Since available peak torque exceeds required torque, the motor satisfies acceleration requirements.

Power required at 80 km/h:

$$P = F \times v$$

Assuming tractive force ≈ 1000 N
 Speed = 22.2 m/s

$$P \approx 22kW (instantaneous peak demand)$$

Continuous power demand during racing $\approx 5\text{--}7$ kW, which is within rated motor power (5 kW) and peak power (9 kW).

1.4 Dynamic Running Conditions

Under acceleration and high-speed operation:

- Peak current draw: 150 A
- RMS current: 120 A
- Peak power: 9 kW
- Motor efficiency: 94%

The motor maintains high efficiency even under dynamic load conditions, minimizing thermal losses and improving battery utilization.

Derating temperature and cutoff temperature are controlled via the controller's thermal protection system.

2. MOTOR CONTROLLER

The motor controller is a 60–72 V programmable PMSM controller.

Parameter	Value
Rated Voltage	60–72 V
Under Voltage Cutoff	40 V
Rated Current	120 A
Peak Current	150 A
Rated Power	5 kW
Peak Power	10 kW
Efficiency	>95%
IP Rating	IP67
Temperature Range	-20°C to 100°C
Cooling	Forced Air

2.1 Control Strategy

The controller uses Field Oriented Control (FOC), which provides:

- Smooth torque output
 - Improved efficiency
 - Better low-speed response
 - Reduced torque ripple
-

2.2 Three Speed Modes

The controller includes a 3-speed switch with programmable limits:

Mode	Max RPM	Max Current	Max Torque
Eco	2000 rpm	70 A	30 Nm
Normal	3000 rpm	120 A	45 Nm
Sport	4200 rpm	150 A	58 Nm

Sport mode enables achieving 80 km/h top speed.

Dynamic programmable parameters include:

- Current limit
 - Acceleration ramp rate
 - Regenerative braking level
 - Speed limiter
-

3. BATTERY PACK

3.1 Battery Type and Configuration

- Type: Lithium-ion
- Nominal Voltage: 60 V
- Capacity: 100 Ah
- Energy Capacity: 6 kWh
- Configuration: 16S × Parallel

Each lithium-ion cell nominal voltage ≈ 3.7 V

16 cells in series provide approximately 60V.

3.2 Charging and Discharging Parameters

Parameter	Value
Discharge C-rate	1C
Charge C-rate	0.5C
Charge Cutoff Voltage	67.2 V
Discharge Cutoff Voltage	48 V
Depth of Discharge (DOD)	80%
State of Health (SOH)	95%

Battery Management System (BMS):

- Type: Smart BMS
- Peak Current: 150 A
- Protection: Over-voltage, under-voltage, overcurrent, thermal protection

SOC is calculated using Coulomb counting method.

3.3 Range Calculation

Total battery energy:

$$Energy = 60V \times 100Ah = 6000Wh = 6kWh$$

Assuming average power consumption = 4 kW

$$Operating Time = \frac{6}{4} = 1.5 \text{ hours}$$

At average speed 45 km/h:

Estimated range $\approx$ 60–65 km

7. SAFETY SYSTEMS

The kart incorporates multiple electrical and operational safety components to ensure protection against short circuits, overcurrent,

thermal faults, and emergency situations. The following safety systems are integrated into the vehicle:

1. MCCB (Molde Case Circuit Breaker)

An MCCB rated for 60V DC system is used as the primary protection device between the battery pack and the motor controller. The MCCB protects the electrical system from overload and short-circuit conditions. It also acts as a manual isolation switch during maintenance. The selected MCCB is rated above the maximum continuous current (120A) and capable of handling peak currents up to 150A.

2. Fuse

A high-speed DC fuse is installed in series with the battery pack to provide fast-acting protection against sudden short circuits. The fuse rating is selected slightly above the maximum operating current to prevent nuisance tripping while ensuring protection under fault conditions.

3. SSR (Solid State Relay)

A DC-rated Solid State Relay is used to control high-current switching electronically. The SSR enables safe and reliable isolation of the traction system without mechanical arcing. It improves system reliability and enhances electrical safety during startup and shutdown.

4. Kill Switch

An emergency kill switch is installed within easy reach of the driver. In case of an emergency, pressing the kill switch immediately disconnects power to the motor controller, stopping the vehicle safely. An external emergency kill switch is also provided for track marshals.

5. BOT (Brake Over-Travel) Switch

The Brake Over-Travel (BOT) switch is integrated into the brake pedal assembly. When excessive brake pedal travel is detected, the BOT switch automatically cuts off motor power to prevent simultaneous acceleration and braking. This ensures enhanced driver safety and compliance with competition regulations.

8. DFMEA AND DVP CHART:

DFMEA (Design Failure Mode and Effects Analysis)

Risk Priority Number is calculated as:

$$RPN = Severity \times Occurrence \times Detection$$

System	Potential Failure Mode	Effect of Failure	S	O	D	RPN	Recommended Action
Motor	Overheating	Reduced efficiency / shutdown	8	4	3	96	Thermal cutoff & cooling
Battery	Overcharge	Cell damage / fire risk	10	2	2	40	Smart BMS protection
Controller	Overcurrent	System failure	8	3	3	72	Current limiting & fuse
Brake	Brake fade	Increased stopping distance	9	3	4	108	Ventilated disc
Chain Drive	Chain breakage	Loss of drive	7	3	4	84	Proper lubrication & tensioner
Electrical Wiring	Short circuit	System shutdown	9	2	3	54	Proper insulation & MCCB

Higher RPN values indicate higher risk and require corrective action.

DVP (Design Validation Plan)

Test Item	Test Objective	Test Method	Acceptance Criteria	Status
Motor Test	Verify rated RPM & torque	Dynamometer testing	3800 rpm, 58 Nm	Passed
Battery Test	Verify capacity	Charge–discharge cycle	6 kWh output	Passed
Controller Protection	Overcurrent validation	Simulated overload	Auto cutoff	Passed
Braking Test	Check stopping distance	Road test	≤ 5 m from 45 km/h	Passed
Thermal Test	Temperature monitoring	Continuous run	Within safe limits	Passed
Electrical Safety	Insulation & leakage test	Megger test	No leakage	Passed

9.CONCLUSION

The design and development of our electric kart successfully demonstrate the integration of mechanical, electrical, braking, safety, ergonomic, and aerodynamic systems into a single high-performance vehicle. Each subsystem was carefully designed based on engineering calculations, simulations, and validation testing to ensure performance, safety, and reliability.

The transmission system was optimized to achieve a maximum speed of 80 km/h while maintaining strong acceleration and torque delivery. The braking system was designed to provide safe stopping distance with adequate thermal stability. The electrical subsystem, consisting of a 5 kW PMSM motor, programmable motor controller, and 60V lithium-ion battery pack, ensures efficient power delivery and dynamic performance. Proper safety components such as MCCB, fuse, SSR, kill switch, BOT switch, and BMS were integrated to enhance operational safety and fault protection.

Ergonomic considerations were implemented based on 95th percentile male anthropometric data to ensure comfortable driving posture and improved vehicle control. The aerodynamic body design and rear spoiler contribute to enhanced stability and professional racing aesthetics. DFMEA and DVP methodologies were adopted to identify potential risks and validate system performance under real operating conditions.

Overall, the developed kart meets performance expectations and competition standards, demonstrating a balanced combination of speed, safety, efficiency, and reliability.

Learning Outcomes from EKVC 2026

Participating in EKVC 2026 provided valuable hands-on engineering experience beyond theoretical knowledge. As budding engineers, we gained practical exposure in complete vehicle design, system integration, and real-world problem solving.

Through this project, we developed strong skills in engineering calculations, component selection, electrical system integration, battery management, safety analysis, and validation testing. We learned the importance of teamwork, time management, documentation, and technical presentation.

The competition enhanced our understanding of electric vehicle technology and sustainable mobility. It strengthened our confidence in applying engineering principles to practical challenges and motivated us to continue innovation in the field of electric mobility.

The overall experience of EKVC 2026 has been a significant step in shaping us into responsible and technically competent engineers.

THANK YOU

